Duration	50 minutes
Submit your answers in the order of questions	2 points penalty for wrong order
Wrong/No Roll Number or Name	Zero points for the Quiz
Wrong/No Instructor Name	2 points will be deducted from the total

Roll Number	EE18B154
Name	Sreehar Sai Rangarathan
Instructor's Name	ANANTH KRISHNAN

For Office Use only [1] :-

Question:	1	2	Total
Points:	6	6	12
Score:			

This exam has 2 questions, for a total of 12 points.

1. (6 points)
Design and draw the circuit implementation of a counter that goes through the following sequence repeatedly
(000) - (100) - (110) - (111) - (101) ...
Use T-Flip flops. Design your circuit in such a manner that, if the machine accidentally goes to an unspecified state, it should get back to the counting sequence.

2. (6 points)

- A vending machine dispenses soft drinks worth Rs. 10.
- Available coins are of Rs. 2 and Rs. 3 denomination.
- The machine accepts 4 coins in sequence, one coin at a time.
- A sensor senses the inserted coin and outputs a bit denoting whether the coin is of denomination Rs. 2 or Rs. 3.
- The machine outputs bit 1 if the total is exactly 10 and outputs a bit 0 otherwise.
- This output bit is used to decide to dispense the soft drink or to return the coins.

Draw the state transition diagram (Mealy or Moore) that takes the sensor bit as input in sequence and outputs a bit to decide whether to dispense the soft drink or not.

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B

Roll No. | G | E | 1 | 8 | B | 1 | 5 | 4 |

Name : Sreehar Sri PANGANDIAN Total No. of Pages

Quiz I ☐ Quiz II/ Mid-Sem ☑ End-Semester ☐ Make-up ☐ Date : 25/3/19

Semester & Degree : II, Elec Engg. Course No. Part :

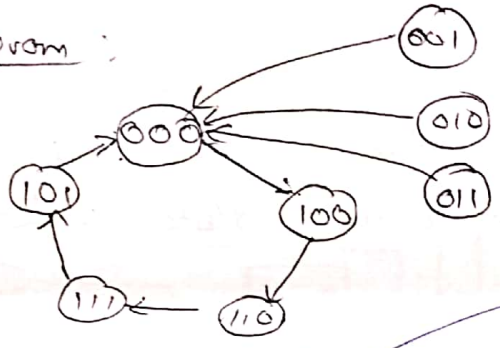
Question No.	1	2	3	4	5	6	7	8	9	10
Marks										

11	12	13	14	15	16	17	18	19	20	Total

Answer on both sides of the paper including the space below

1. Counter (sequence given)

State diagram :



Triggered

State Table

Present State Q_A Q_B Q_C	Next State Q_A^+ Q_B^+ Q_C^+	FF inputs
0 0 0		
0 0 1		
1 1 0		
1 1 1		
1 0 1		

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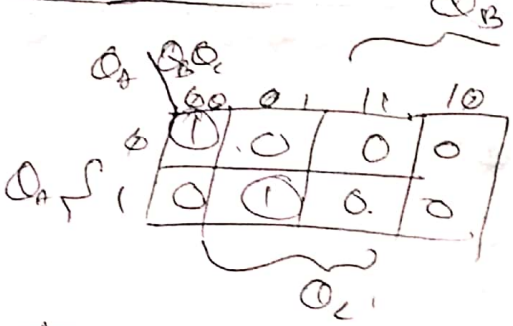
Present State			Next State			Flip-Flop inputs		
Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	T_A	T_B	T_C
0	0	0	1	0	0	1	0	0
0	0	1	x(0)	x(0)	x(0)	0	0	1
0	1	0	x(0)	x(0)	x(0)	0	1	0
0	1	1	x(0)	x(0)	x(0)	0	1	1
1	0	0	1	1	0	0	1	0
1	0	1	0	0	0	1	0	1
1	1	0	1	1	1	0	0	1
1	1	1	1	0	1	0	1	0

Assigning Don't cares to be equal to 000 so that an unused state is fed back into the sequence.

→ Using 3T-flip flops. (Directly acts as o/p)

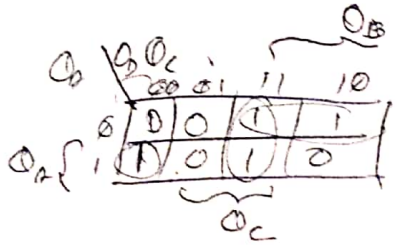
Simplification for feedback circuit. (Flip-Flop inputs)

T_A :



$$T_A = Q_A' Q_B' Q_C' + Q_A Q_C Q_B'$$

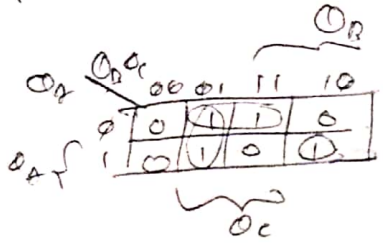
T_B :



$$T_B = Q_B Q_C + Q_A' Q_B + Q_A Q_B' Q_C'$$

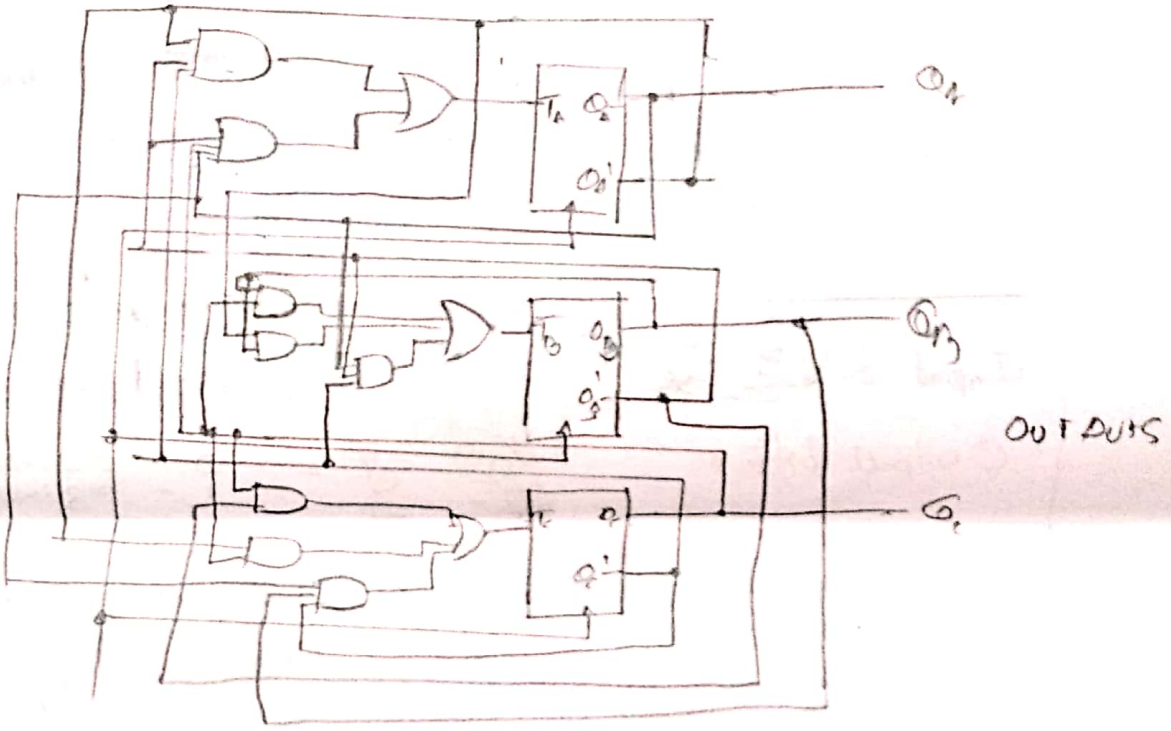
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T_C - :



$$T_C = Q_B' Q_C + Q_A' Q_C + Q_A Q_B Q_C'$$

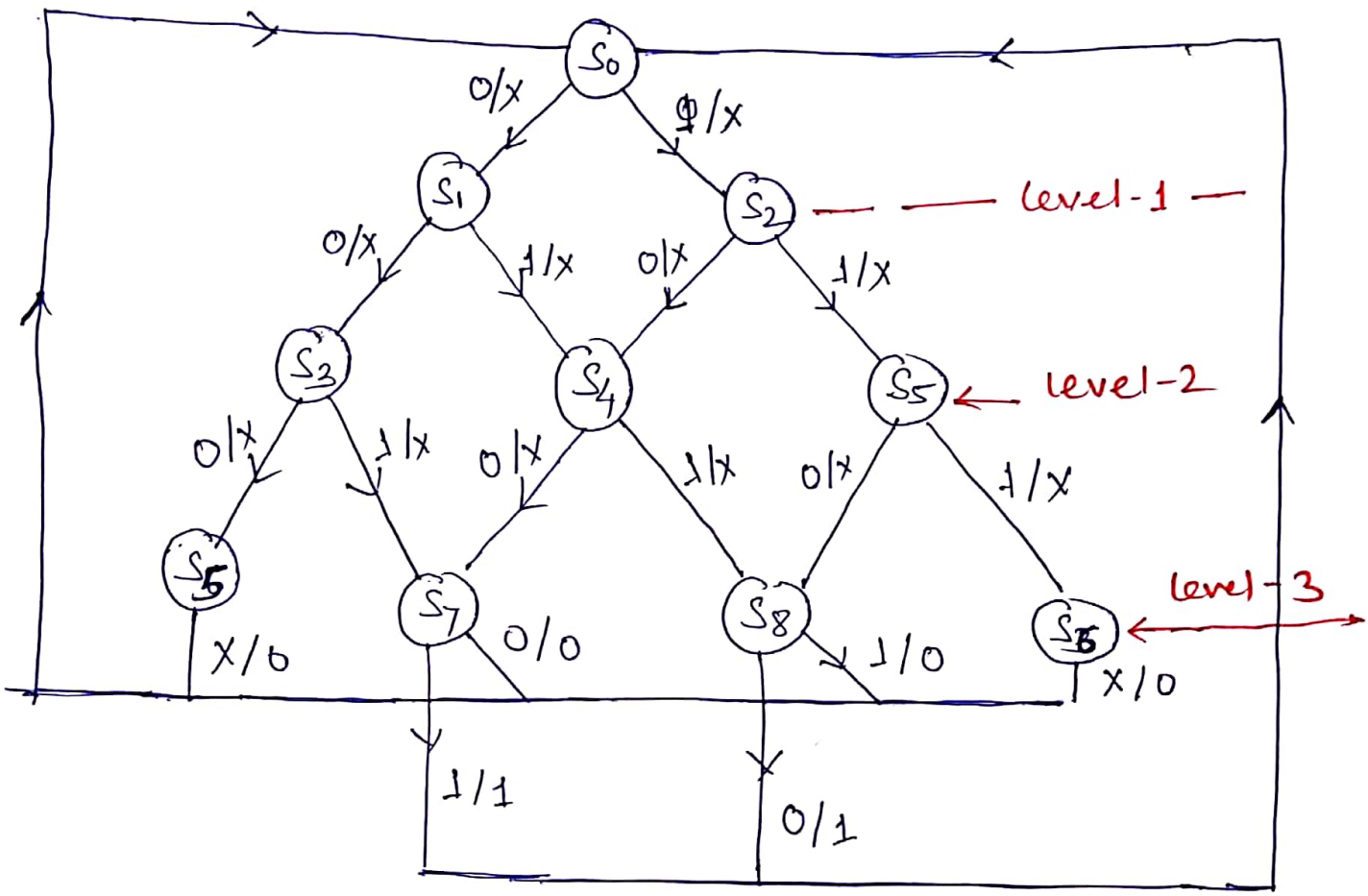
Logic diagram :



Output is same as T-flip flop outputs

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Q-2 Solution:



Marking:-

(1) level 1, drawn with 3 states — 1 Marks

(2) level-2, drawn with 3 states — 2 Marks
drawn with More then 3 states — 1 Marks

(3) level-3, drawn with 3 States — 2 Marks
drawn with 4 States — 1½ Marks
drawn with 6 States — 1 Marks
drawn with More then 6 — ½ Marks

(4) Correct output, terminated at S0 — 1 Marks.

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